

the question of how many forward kinematics solutions can exist for the general 6–6 platform consisting of six SPS legs (with the prismatic joints actuated) connecting a fixed platform to a moving platform. Raghavan and Roth [143] showed that there can be at most 40 solutions, while Husty [62] developed an algorithm for finding all 40 solutions.

Singularities of closed chains have also received considerable attention in the literature. The terminology for closed-chain singularities used in this chapter was introduced in [136]; in particular, the distinction between degenerate and nondegenerate actuator singularities derives in part from similar terminology used in Morse theory to identify those critical points where the Hessian is singular (i.e., degenerate). The $3\times\text{UPU}$ mechanism, which is addressed in the exercises in both Chapter 2 and the current chapter, can exhibit rather unusual singularity behavior; a more detailed singularity analysis of this mechanism can be found in [52, 35].

7.6 Exercises

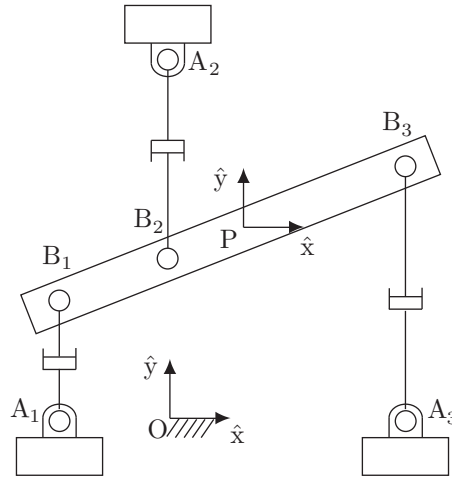


Figure 7.10: $3\times\text{RPR}$ planar parallel mechanism.

Exercise 7.1 In the $3\times\text{RPR}$ planar parallel mechanism of Figure 7.10 the prismatic joints are actuated. Define $a_i \in \mathbb{R}^2$ to be the vector from the fixed-

frame origin O to joint A_i , $i = 1, 2, 3$, expressed in fixed-frame coordinates. Define $b_i \in \mathbb{R}^2$ to be the vector from the moving-platform-frame origin P to joint B_i , $i = 1, 2, 3$, defined in terms of the moving-platform-frame coordinates.

- Solve the inverse kinematics.
- Derive a procedure to solve the forward kinematics.
- Is the configuration shown in the figure an end-effector singularity? Explain your answer by examining the inverse kinematics Jacobian. Is this also an actuator singularity?

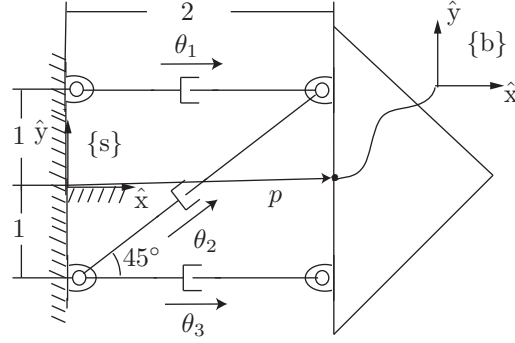
Exercise 7.2 For the $3 \times \text{RPR}$ planar parallel mechanism in Figure 7.11(a), let ϕ be the angle measured from the $\{s\}$ -frame $\hat{x}$ -axis to the $\{b\}$ -frame $\hat{x}$ -axis, and $p \in \mathbb{R}^2$ be the vector from the $\{s\}$ -frame origin to the $\{b\}$ -frame origin, expressed in $\{s\}$ -frame coordinates. Let $a_i \in \mathbb{R}^2$ be the vector from the $\{s\}$ -frame origin to the three joints fixed to ground, $i = 1, 2, 3$ (note that two of the joints are overlapping), expressed in $\{s\}$ -frame coordinates. Let $b_i \in \mathbb{R}^2$ be the vector from the $\{b\}$ -frame origin to the three joints attached to the moving platform, $i = 1, 2, 3$ (note that two of the joints are overlapping), expressed in $\{b\}$ -frame coordinates. The three prismatic joints are actuated, and the leg lengths are θ_1 , θ_2 , and θ_3 , as shown.

- Derive a set of independent equations relating (ϕ, p) and $(\theta_1, \theta_2, \theta_3)$.
- What is the maximum possible number of forward kinematics solutions?
- Assuming static equilibrium, given joint forces $\tau = (1, 0, -1)$ applied at joints $(\theta_1, \theta_2, \theta_3)$, find the planar wrench (m_{bz}, f_{bx}, f_{by}) in the end-effector frame $\{b\}$.
- Now construct a mechanism with three connected $3 \times \text{RPR}$ parallel mechanisms as shown in Figure 7.11(b). How many degrees of freedom does this mechanism have?

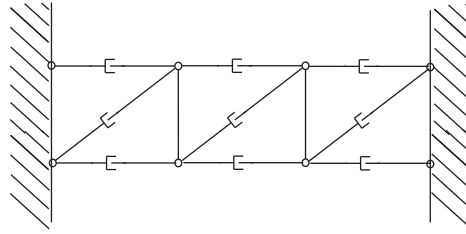
Exercise 7.3 For the $3 \times \text{RRR}$ planar parallel mechanism shown in Figure 7.12, let ϕ be the orientation of the end-effector frame and $p \in \mathbb{R}^2$ be the vector p expressed in fixed-frame coordinates. Let $a_i \in \mathbb{R}^2$ be the vector a_i expressed in fixed-frame coordinates and $b_i \in \mathbb{R}^2$ be the vector b_i expressed in the moving body-frame coordinates.

- Derive a set of independent equations relating (ϕ, p) and $(\theta_1, \theta_2, \theta_3)$.
- What is the maximum possible number of inverse and forward kinematic solutions for this mechanism?

Exercise 7.4 Figure 7.13 shows a six-bar linkage in its zero position. Let (p_x, p_y) be the position of the $\{b\}$ -frame origin expressed in $\{s\}$ -frame coordinates,



(a) 3×RPR planar parallel mechanism.



(b) Truss.

Figure 7.11: 3×RPR planar parallel mechanism and truss structure.

and let ϕ be the orientation of the $\{b\}$ frame. The inverse kinematics problem is defined as that of finding the joint variables (θ, ψ) given (p_x, p_y, ϕ) .

- In order to solve the inverse kinematics problem, how many equations are needed? Derive these equations.
- Assume that joints A, D, and E are actuated. Determine whether the configuration shown in Figure 7.13 is an actuator singularity by analyzing an equation of the form

$$\begin{bmatrix} H_a & H_p \end{bmatrix} \begin{bmatrix} \dot{q}_a \\ \dot{q}_p \end{bmatrix} = 0,$$

where q_a is the vector of the actuated joints and q_p is the vector of the passive joints.

- Suppose instead that joints A, B, and D are actuated. Find the forward kinematics Jacobian J_a from $\mathcal{V}_s = J_a \dot{q}_a$, where $\mathcal{V}_s$ is the twist in $\{s\}$ -frame coordinates and $\dot{q}_a$ is the vector of actuated joint rates.

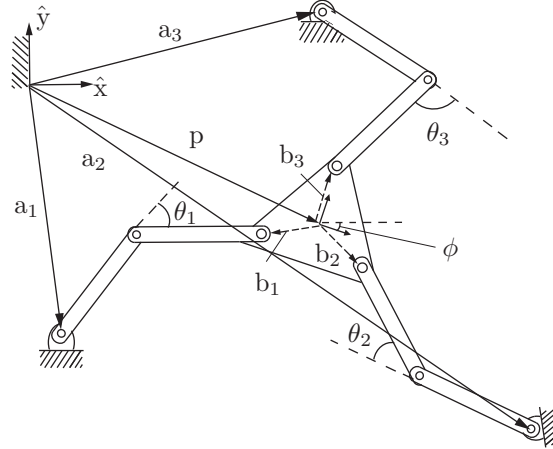


Figure 7.12: 3×RRR planar parallel mechanism.

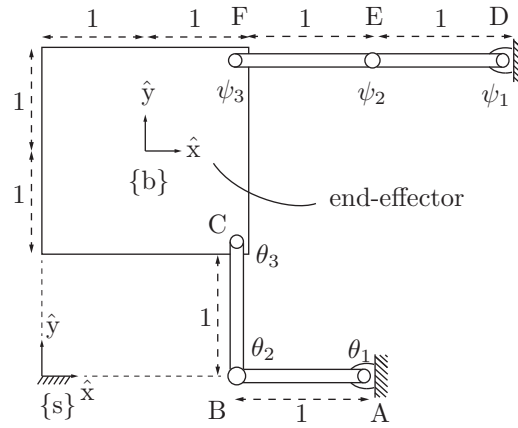


Figure 7.13: A six-bar-linkage.

Exercise 7.5 Consider the 3×PSP spatial parallel mechanism of Figure 7.14.

- How many degrees of freedom does this mechanism have?
- Let $R_{sb} = \text{Rot}(\hat{z}, \theta)\text{Rot}(\hat{y}, \phi)\text{Rot}(\hat{x}, \psi)$ be the orientation of the body frame

$\{b\}$, and let $p_{sb} = (x, y, z) \in \mathbb{R}^3$ be the vector from the $\{s\}$ -frame origin to the $\{b\}$ -frame origin (both R_{sb} and p_{sb} are expressed in $\{s\}$ -frame coordinates). The vectors $a_i, b_i, d_i, i = 1, 2, 3$, are defined as shown in the figure. Derive a set of independent kinematic constraint equations relating $(\theta, \phi, \psi, x, y, z)$ and the defined vectors.

- (c) Given values for (x, y, z) , is it possible to solve for the vertical prismatic joint values s_i , where $s_i = \|d_i\|$ for $i = 1, 2, 3$? If so, derive an algorithm for doing so.

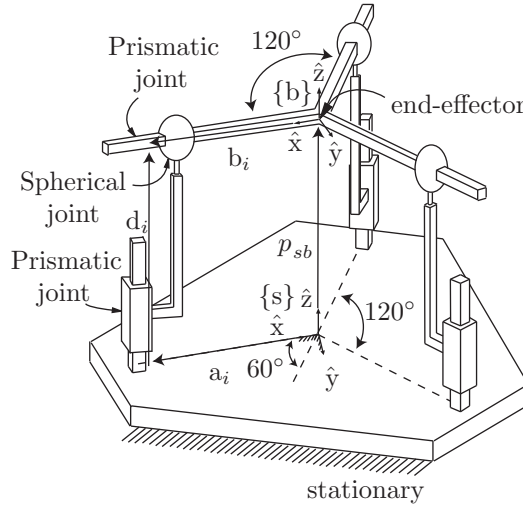


Figure 7.14: 3×PSP spatial parallel manipulator.

Exercise 7.6 The Eclipse mechanism of Figure 7.15 is a six-dof parallel mechanism whose moving platform can tilt by $\pm 90^\circ$ with respect to ground and can also rotate by 360° about the vertical axis. Assume that the six sliding joints are actuated.

- Derive the forward and inverse kinematics. How many forward kinematic solutions are there for general nonsingular configurations?
- Find and classify all singularities of this mechanism.

Exercise 7.7 For the Delta robot of Figure 7.1(b), obtain the following:

- the forward kinematics,
- the inverse kinematics,

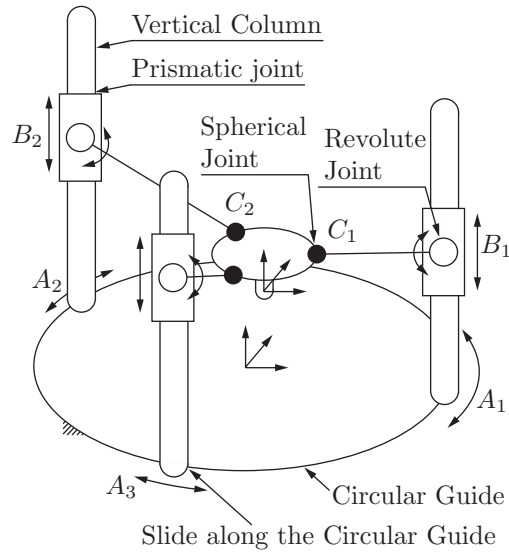


Figure 7.15: The Eclipse mechanism.

- (c) the Jacobian J_a (assume that the revolute joints at the fixed base are actuated).
- (d) Identify all actuator singularities of the Delta robot.

Exercise 7.8 In the $3 \times \text{UPU}$ platform of Figure 7.16, the axes of the universal joints are attached to the fixed and moving platforms in the sequence indicated, i.e., axis 1 is attached orthogonally to the fixed base, while axis 4 is attached orthogonally to the moving platform. Obtain the following:

- (a) the forward kinematics,
- (b) the inverse kinematics,
- (c) the Jacobian J_a (assume that the revolute joints at the fixed base are actuated).
- (d) Identify all actuator singularities of this robot.
- (e) If you can, build a mechanical prototype and see whether the mechanism behaves as predicted by your analysis.

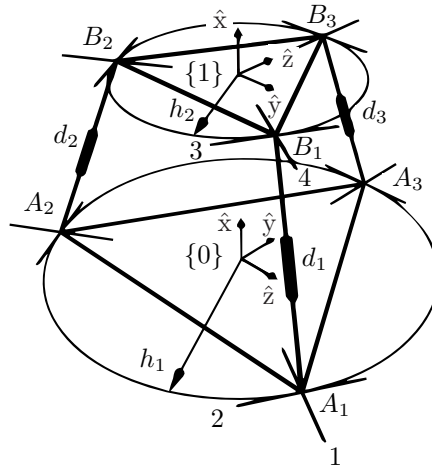


Figure 7.16: The $3 \times \text{UPU}$ mechanism.

